



HTML + CSS

Web Application Development

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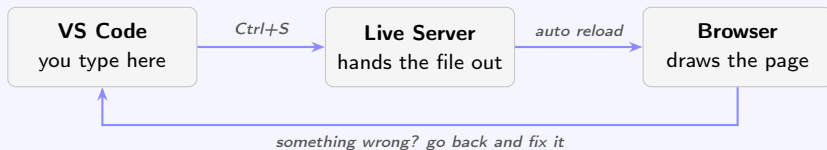
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1. Tools and Workflow

Your Toolbox: Editor, Live Server, Browser

The loop you repeat every few seconds



Why not just double-click the file?

- Double-click opens `file:///...` - works, but not a server
- Live Server gives `http://localhost:5500` - a real web address

What you do NOT need

- No compiler, no build step
- No paid software
- Any text editor works - VS Code is just easier

Set Up in Five Minutes

1. Install VS Code
code.visualstudio.com

2. Add Live Server
Ctrl+Shift+X → Install

3. Create index.html
inside a new folder

4. Click Go Live
bottom right

Go Live - bottom right of the status bar

```
8  </head>
9  <body>
10 |   <h1>Enable Live Server </h1>
11  </body>
12  </html>
```



Ln 12, Col 8 Spaces: 4 UTF-8 CRLF HTML Go Live Prettier

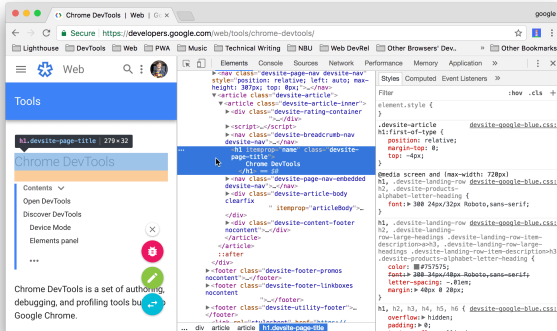
DevTools: Look Inside Any Page

F12
opens DevTools

Elements
live page structure

Ctrl+U
raw text from server

Hover a line
that part lights up



2. What HTML Is

A Web Page Is Text With Tags

What you write - plain text

```
<h1>Pho Thin</h1>
<p>Beef noodle soup
  60,000 VND</p>
<a href="/order">Order now</a>
```

tag names in blue, attribute in orange

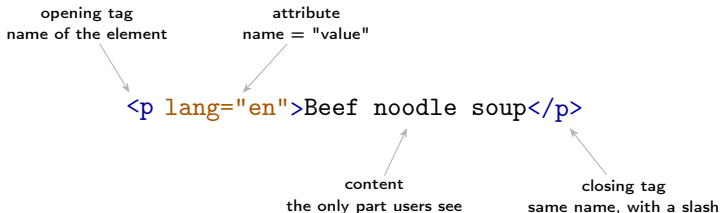
What the browser shows



- Characters only - no pixels, no colours
- Words in angle brackets = tags
- A tag opens <p> and closes </p>

- Browser reads the tags, then draws
- <h1> = heading, <p> = paragraph
- <a> = link the user can click

Parts of a Tag



Rules to remember

- Element = opening tag + content + closing tag
- Attribute value goes in double quotes
- Attributes live in the opening tag only
- One tag can carry several attributes

What ends up on screen

- Only Beef noodle soup
- Tag names never show
- `lang="en"` never shows - it tells the browser the language

Tags That Close, Tags That Don't

Most tags come in pairs

- They wrap content: `<p>Beef noodle soup</p>`
- `<html> <head> <title> <body> <h1> <p> <a> <ul> <ol> <li> <div> <table>`
- Closing tag = same name with a slash in front

A few never close

- Nothing goes inside them, so there is nothing to wrap
- `<br> <hr> <img> <input> <meta>`
- Older pages write `<br />` - same meaning

Forget one closing tag, and the rest of the page pays for it

```
<a href="/order">Order now  no </a> here  
<p>Beef noodle soup 60,000 VND</p>
```

Result: the price line turns blue and clickable too - the link never ended.

The Page Skeleton

Every file starts like this

```
<!DOCTYPE html>
<html>
  <head>
    <title>Pho Thin</title>
  </head>
  <body>
    <h1>Pho Thin</h1>
  </body>
</html>
```

Boxes inside boxes

```
<html>
  <head>
    title of the tab
  </head>
  <body>
    everything the user sees
    headings, text, images, links
  </body>
</html>
```

Points to make

- `<!DOCTYPE html>` is not a tag - it tells the browser to read the file as modern HTML
- `<head>` holds information about the page - `<title>` `<meta>` `<style>` `<script>` - none of it is drawn
- `<title>` is the tab name, the name your bookmark gets, and the line Google shows in its results
- `<body>` holds the page itself; from here on, every tag we meet goes inside it
- The indenting is only for you - the browser ignores spaces and line breaks

Text Tags

What you write

```
<h1>Pho Thin</h1>
<h2>Menu</h2>
<p>Beef noodle soup
  <strong>60,000 VND</strong>
  <em>herbs included</em></p>
<hr>
<p>Open 6am - 10am<br>
  Closed on Monday</p>
```

What the browser shows

Pho Thin

Menu

Beef noodle soup 60,000 VND *herbs included*

Open 6am - 10am
Closed on Monday

Points to make

- Six heading levels, <h1> down to <h6> - <h1> is the biggest
- makes text bold, makes it italic
-
 breaks a line, <hr> draws a rule - neither one closes

List Tags

What you write

```
<ul>
  <li>Beef</li>
  <li>Chicken</li>
</ul>
<ol>
  <li>Pick a bowl</li>
  <li>Pay at the counter</li>
</ol>
<dl>
  <dt>Pho</dt>
  <dd>Rice noodle soup</dd>
</dl>
```

What the browser shows

- Beef
 - Chicken
1. Pick a bowl
 2. Pay at the counter

Pho

Rice noodle soup

Points to make

- `<li>` is one item - it works inside both `<ul>` and `<ol>`
- `<ul>` gives bullets, `<ol>` counts 1, 2, 3 - the browser writes the numbers, not you
- `<dl>` comes in pairs: `<dt>` is the word, `<dd>` is what it means

Link and Image Tags

What you write

```
<a href="https://phothin.vn">Visit us</a>



<a href="/order">
  
</a>
```

What the browser shows

[Visit us](#)



the framed one is a link

Writing the address

- Same folder: `src="bowl.jpg"`
- Sub-folder: `src="figures/bowl.jpg"`
- Parent folder: `href="../index.html"`
- Full address: protocol + domain + path
- No file name? The server sends `index.html`

Choosing the picture

- JPEG - photos, small file
- PNG - see-through background
- SVG - logos and icons, sharp at any size
- GIF - short animations only
- alt - text if the picture fails to load

Table Tags

What you write

```
<table>
  <tr><th>Dish</th><th>Price</th></tr>
  <tr><td>Pho bo</td><td>60,000</td></tr>
  <tr><td>Pho ga</td><td>55,000</td></tr>
</table>
```

What the browser really shows

Dish	Price
Pho bo	60,000
Pho ga	55,000

Points to make

- `<tr>` is one row, `<td>` is one cell, `<th>` is a heading cell - bold and centred
- Cells go left to right, rows top to bottom; the browser lines the columns up for you

Table Tags

What you write

```
<table>
  <tr><th>Dish</th><th>Price</th></tr>
  <tr><td>Pho bo</td><td>60,000</td></tr>
  <tr><td>Pho ga</td><td>55,000</td></tr>
</table>
```

What the browser really shows

Dish	Price
Pho bo	60,000
Pho ga	55,000

Points to make

- `<tr>` is one row, `<td>` is one cell, `<th>` is a heading cell - bold and centred
- Cells go left to right, rows top to bottom; the browser lines the columns up for you

A table has no lines of its own

- Do not expect a grid - HTML draws no borders at all
- Lines come from CSS: `table, th, td { border: 1px solid; }`
- Add `border-collapse: collapse;` or every line comes out double

Cells That Span

What you write

```
<style>
  table { border-collapse: collapse; }
  table, th, td { border: 1px solid; }
</style>

<table>
  <tr><th colspan="2">Menu</th></tr>
  <tr><td rowspan="2">Noodles</td>
    <td>Pho bo</td></tr>
  <tr><td>Pho ga</td></tr>
</table>
```

What the browser shows

Menu	
Noodles	Pho bo
	Pho ga

Points to make

- `colspan="2"` - one cell takes the place of two sitting side by side
- `rowspan="2"` - one cell takes the place of two stacked on top of each other
- Count your cells: a spanned cell is written once, so the row below it has one `<td>` fewer

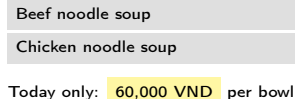
Block and Inline

What you write

```
<div>Beef noodle soup</div>
<div>Chicken noodle soup</div>

<p>Today only: <span>60,000 VND</span> per
bowl</p>
```

Shaded to show the space taken



Beef noodle soup

Chicken noodle soup

Today only: 60,000 VND per bowl

Block-level

- Starts on a new line and takes the full width
- `<h1>`-`<h6>` `<p>` `<ul>` `<ol>` `<li>` `<div>` `<table>`
- `<header>` `<nav>` `<main>` `<footer>` behave like `<div>` but say what the part is for

Inline

- Stays in the line and takes only the width of its content
- `<a>` `<img>` `<span>` `<strong>` `<em>` `<br>`

Points to make

- This is why `<h1>` always jumps to a new line while `<a>` stays inside the sentence
- `<div>` and `<span>` are empty boxes with no look of their own - they group things so that CSS can style them later

Exercise: Build This Page

The page you must build

Pho Thin

Since 1979

Our bowls

- Pho bo
- Pho ga
- Pho tai

Prices

Pho bo	60,000
Pho ga	55,000

[Order now](#)

[photo of a bowl]

Rules

1. Start from the skeleton - the browser tab must say Pho Thin
2. One <h1>, one <h2>
3. Since 1979 in italic, the prices in bold
4. The three bowls as a bulleted list
5. In the table, the Prices row covers both columns
6. Order now links to <https://phothin.vn>
7. One picture, with an alt that describes it - a local file or a web address
8. Done early? Add a <dl> for tai and nam, and make one cell cover two rows

Two questions when you are done

1. Which tags on your page need no closing tag, and why?
2. Pho Thin shows up in the tab and in the page - which tag gives each one?

3. Styling with CSS

From Tags to Looks

The HTML never changes

```
<h1>Pho Thin</h1>
<p class="price">60,000 VND</p>
```

No CSS

Pho Thin

60,000 VND

With three lines of CSS

```
h1 { color: #b00000; text-align: center; }
.price { background: yellow; }
```

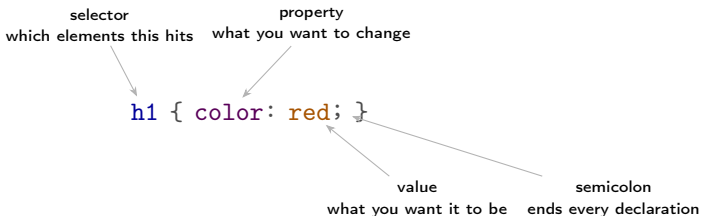
Pho Thin

60,000 VND

Points to make

- HTML says what a thing is; CSS says how it looks
- Same page, different style sheet - the words on screen never moved
- This is why we spent time on `class` and `<div>`: CSS needs something to aim at

Parts of a CSS Rule



Rules to remember

- One declaration = property: value;
- One rule can hold as many declarations as you like
- The curly braces mark where the rule starts and stops
- Forget a semicolon and the next line is ignored

Three declarations in one rule

```
h1 {  
  color: red;  
  text-align: center;  
  font-size: 32px;  
}
```

Three Ways to Attach CSS

1. On the tag

```
<h1 style=
  "color: red;">
Pho Thin</h1>
```

one tag only

2. In the head

```
<head>
<style>
  h1 { color: red; }
</style>
</head>
```

one page only

3. In its own file

```
<link rel="stylesheet"
      href="style.css">
```

style.css:

```
h1 { color: red; }
```

every page you link it to

Points to make

- When two rules clash, `style="..."` on the tag always wins
- Between `<style>` and the linked file, whichever comes later in the `<head>` wins
- Use the separate file: change one line, every page that links it changes

Selectors: Class, ID, type

What you write

```
<h1 id="name">Pho Thin</h1>
<p>Beef noodle soup</p>
<p class="price">60,000 VND</p>
<p class="price">55,000 VND</p>

h1, h2 { font-family: sans-serif; }
p { color: gray; }
.price { font-weight: bold; }
#name { color: red; }
```

What the browser shows

Pho Thin

Beef noodle soup

60,000 VND

55,000 VND

Points to make

- p - just the tag name, hits every paragraph on the page
- .price - a dot in front; the same class can sit on as many elements as you want
- #name - a hash in front; an id belongs to one element only
- h1, h2 - a comma applies the same rule to both tags
- Two rules can hit the same element: the prices are grey *and* bold

Which Rule Wins

Check in this order

1. More specific selector wins: `id > class > tag`
2. Still tied? The one written later wins
3. `style="..."` on the tag beats every selector

Two traps

- One `id` beats any number of classes - the counts do not add up
- `!important` overrules all of it. You will meet it in other people's code; do not write it yourself

Class beats tag

```
.price { color: red; }  
p { color: gray; }
```

`<p class="price">` comes out **red** - class beats tag even though `p` is written later

Last line wins

```
p { color: red; }  
p { color: gray; }
```

`<p>` comes out **grey** - both are one tag, so the last line wins

When your CSS does nothing

- Press **F12**, click the element, read the Styles panel - the rule that lost is shown crossed out



Colours and Backgrounds

The HTML

```
<h1>Pho Thin</h1>
<p class="price">60,000 VND</p>
<p class="note">Open 6am - 10am</p>
<p class="badge">Best seller</p>
```

The CSS

```
h1 { color: darkred; }
.price { color: #b00000; }
.note { color: rgb(90, 90, 90); }
.badge { background-color: gold; }
```

What the browser shows

Pho Thin

60,000 VND

Open 6am - 10am

Best seller

Points to make

- color paints the text, background-color paints the box behind it
- red = #ff0000 = rgb(255, 0, 0) - the same colour written three ways
- Hex is three pairs - red, green, blue - each from 00 to ff



Colour Values

Names

	red		tomato
	gold		olive
	teal		steelblue
	indigo		gray
	white		black

Hex - #rrggbb

	#b00000
	#00a86b
	#1e90ff
	#ffa500
	#333333
	#ff00 short form of #ff0000

rgb and rgba

	rgb(255, 99, 71)
	rgb(0, 128, 128)
	rgb(124, 252, 0)
	rgba(0, 0, 0, 0.5)
	rgba(255, 215, 0, 0.3)

Each number 0 to 255. The fourth one is opacity: 1 solid, 0 invisible.

hsl

	hsl(0, 100%, 50%)
	hsl(120, 100%, 50%)
	hsl(240, 100%, 50%)
	hsl(0, 100%, 80%)
	hsl(0, 50%, 50%)

Hue 0 to 360 around the colour wheel, then how strong, then how light.

Fonts

The HTML

```
<h1>Pho Thin</h1>
<p>Beef noodle soup</p>
<p class="note">herbs included</p>
```

The CSS

```
body { font-family: Arial, Helvetica, sans-serif; }
h1 { font-size: 32px; font-weight: bold; }
.note { font-style: italic; font-size: 12px; }
```

What the browser shows

Pho Thin

Beef noodle soup

herbs included

The three families

Pho Thin - serif

Pho Thin - sans-serif

Pho Thin - monospace

Points to make

- The list is read left to right - the browser takes the first font the computer actually has
- The last name must be serif, sans-serif or monospace - every computer has one of those
- font-weight: bold looks like but carries no meaning - keep meaning in HTML, looks in CSS

Choosing a Font

Explanation of the list

`font-family: Arial, Helvetica, sans-serif;`

Arial installed? → use Arial

No? Helvetica installed? → use Helvetica

Neither? → use its own sans-serif

Bold two ways - pick the right one

- `<strong>...</strong>` - HTML. Says the words matter
- `font-weight: bold` - CSS. Says nothing; a screen reader will not stress it
- Important to the reader → `<strong>`. Bold only because it looks better → CSS

Generic Font Name	Type of Font	Example
Serif	Fonts with serifs	Times
Sans-serif	Fonts without serifs	Arial
Monospace	Fixed-width fonts	Courier
Cursive	Fonts that emulate handwriting	Comic Sans
Fantasy	Decorative fonts for titles, and so on	Impact

Font Values

font-weight

Pho Thin	normal
Pho Thin	bold
Pho Thin	100 200 300 400
Pho Thin	500 600 700 800 900
	lighter bolder

font-style

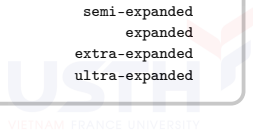
Pho Thin	normal
<i>Pho Thin</i>	italic
<i>Pho Thin</i>	oblique

font-size

Pho Thin	12px
Pho Thin	16px
Pho Thin	32px
	px pt % em rem
	small medium
	large

font-stretch

narrower letters	ultra-condensed
	extra-condensed
	condensed
	semi-condensed
default	normal
wider letters	semi-expanded
	expanded
	extra-expanded
	ultra-expanded



Formatting Text

The HTML

```
<h1>Pho Thin</h1>
<p>Beef noodle soup with fresh herbs</p>
<p><span class="old">70,000 VND</span> 60,000 VND</p>
<a href="/order">Order now</a>
```

What you write

```
h1 { text-align: center; }
p { line-height: 1.6; }
a { text-decoration: none; }
.old { text-decoration: line-through; }
```

What the browser shows

Pho Thin

Beef noodle soup with fresh herbs

~~70,000 VND~~ 60,000 VND

[Order now](#)

Points to make

- text-align: center centres the text *inside* its box - it does not move the box itself
- text-decoration: none is how you take the underline off a link; line-through is how you show an old price
- line-height: 1.6 means 1.6 times the font size

Text Values

text-align

Pho Thin	left
Pho Thin	center
Pho Thin	right
Pho Thin Hanoi	justify

text-decoration

60,000 VND	none
<u>60,000 VND</u>	underline
<u>60,000 VND</u>	overline
60,000 VND	line-through

vertical-align - inline

Pho Thin	baseline
Pho Thin	sub
Pho Thin	super

vertical-align - more values

Pho Thin	text-top
Pho Thin	text-bottom
in table cells	top middle bottom

Pseudo-classes

a:link - never opened yet

```
a:link { color: red; }
```

Order now

a:visited - opened before

```
a:visited { color: green; }
```

Order now

a:hover - mouse is on it

```
a:hover { color: deeppink; }
```

Order now 

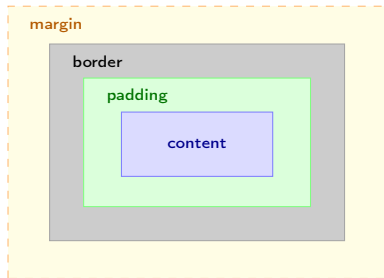
a:active - button held down

```
a:active { background: yellow; }
```

Order now 

The Box Model

Every element is four boxes



The CSS behind it

```
.card {  
  width: 200px;  
  padding: 20px;  
  border: 5px solid;  
  margin: 10px;  
}
```

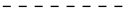
content	200
padding left+right	20 + 20
border left+right	5 + 5
box drawn	250px
margin left+right	10 + 10
space used	270px

Points to make

- padding is inside the border, margin is outside it - padding takes the background colour, margin never does
- width sets the content only; padding and border are added on top of it
- Press F12 and hover an element - DevTools draws these four boxes in the same four colours

Box Values

border-style

	solid
	dashed
	dotted
	double
	none

border-width

	thin	1px
	medium	3px
	thick	5px

margin and padding shorthand

margin: 10px
all four sides

margin: 10px 20px
top+bottom, left+right

margin: 10px 20px 30px
top, left+right, bottom

margin: 10px 20px 30px 40px
top, right, bottom, left - clockwise

margin: 0 auto
centres the box

width and height

a fixed size	200px
share of the parent	50%
browser decides	auto
never wider than	max-width
never shorter than	min-height



Sizing the Box

Same width - two results

```
.a { width: 200px; padding: 20px; border: 5px solid; }  
.b { width: 200px; padding: 20px; border: 5px solid;  
    box-sizing: border-box; }
```

What you get

.a drawn 250px wide



.b drawn 200px wide



Points to make

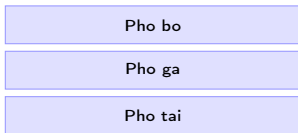
- Default is content-box: width counts the content only, so padding and border push the box wider
- border-box: width is the final width, and padding and border eat into it - the content shrinks instead
- Real projects start the stylesheet with `* { box-sizing: border-box; }` so every box behaves the second way

Boxes in a Row

The HTML

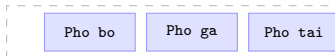
```
<div class="menu">
  <div class="card">Pho bo</div> <div class="card">Pho ga</div> <div class="card">Pho tai</div>
</div>
```

No CSS - div is block



With three lines of CSS

```
.menu { display: flex; gap: 10px;
        justify-content: center; }
```



Points to make

- `display: flex` goes on the parent; the children line up in a row by themselves
- `gap` sets the space between children - no margin tricks needed
- `justify-content: flex-start center space-between space-around` - where the row sits
- `align-items: center` centres them top to bottom - the thing that used to be hard

Styling a Table

The CSS

```
table { border-collapse: collapse; }  
th, td { border: 1px solid #999;  
  padding: 8px; text-align: left; }  
th { background-color: #b00000; color: white; }  
tr:hover { background-color: #fff3cd; }
```

What the browser shows

Dish	Price
Pho bo	60,000
Pho ga	55,000
Pho tai	65,000

the yellow row is under the mouse

Points to make

- This is the promise from the HTML section: the lines were never in the table, they come from CSS
- `border-collapse: collapse` merges the doubled lines into one
- Style `th`, `td` together first, then override `th` on its own
- `tr:hover` is a **pseudo-class** - it only applies while the mouse is over that row

Table Values

border-collapse

Pho bo	60
Pho ga	55

collapse one line

Pho bo	60
Pho ga	55

separate gaps

text-align in a cell

Pho bo	left
Pho bo	center
Pho bo	right

vertical-align in a cell

Pho	top
Pho	middle
Pho	bottom

size and caption

fills the parent	width: 100%
fits the content	width: auto
gap between cells	border-spacing
where the caption goes	caption-side
above or below	top bottom



Exercise

Read this page

```
<h1 class="title">Pho Thin</h1>
<p class="price">60,000 VND</p>

title { color: red; }
.price { color: green
         font-size: 20px; }

.card { width: 300px; padding: 10px;
        border: 2px solid; }
```

Four questions

1. What colour is the heading, and why is it not red?
2. One character is missing in the second rule. Which one?
3. Because of that, will the price be green? Will it be 20px?
4. How wide is .card on screen?

Thank you for listening!